

ThaiGrid and E-science in Thailand

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ThaiGrid

- A partnership project to explore grid computing technology and application in Thailand. Project started since December 2000
- Link: <http://www.thaigrid.net>
- Currently funded by
 - National Research Council of Thailand (NRCT) 1.0 Million
 - Commission on Higher Education, Ministry of Education (2.6 Million)
- Infrastructure funded by
 - KU, KMITNB, SUT
 - NTL NECTEC
- Contact
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Members

● 7 universities

- Kasetsart University
- King Mongkut's Institute of Technology North Bangkok
- Suranaree University of Technology
- Asian Institute of Technology
- Chulalongkorn University
- Walailak University
- Chiangmai University

● 1 Government Agency

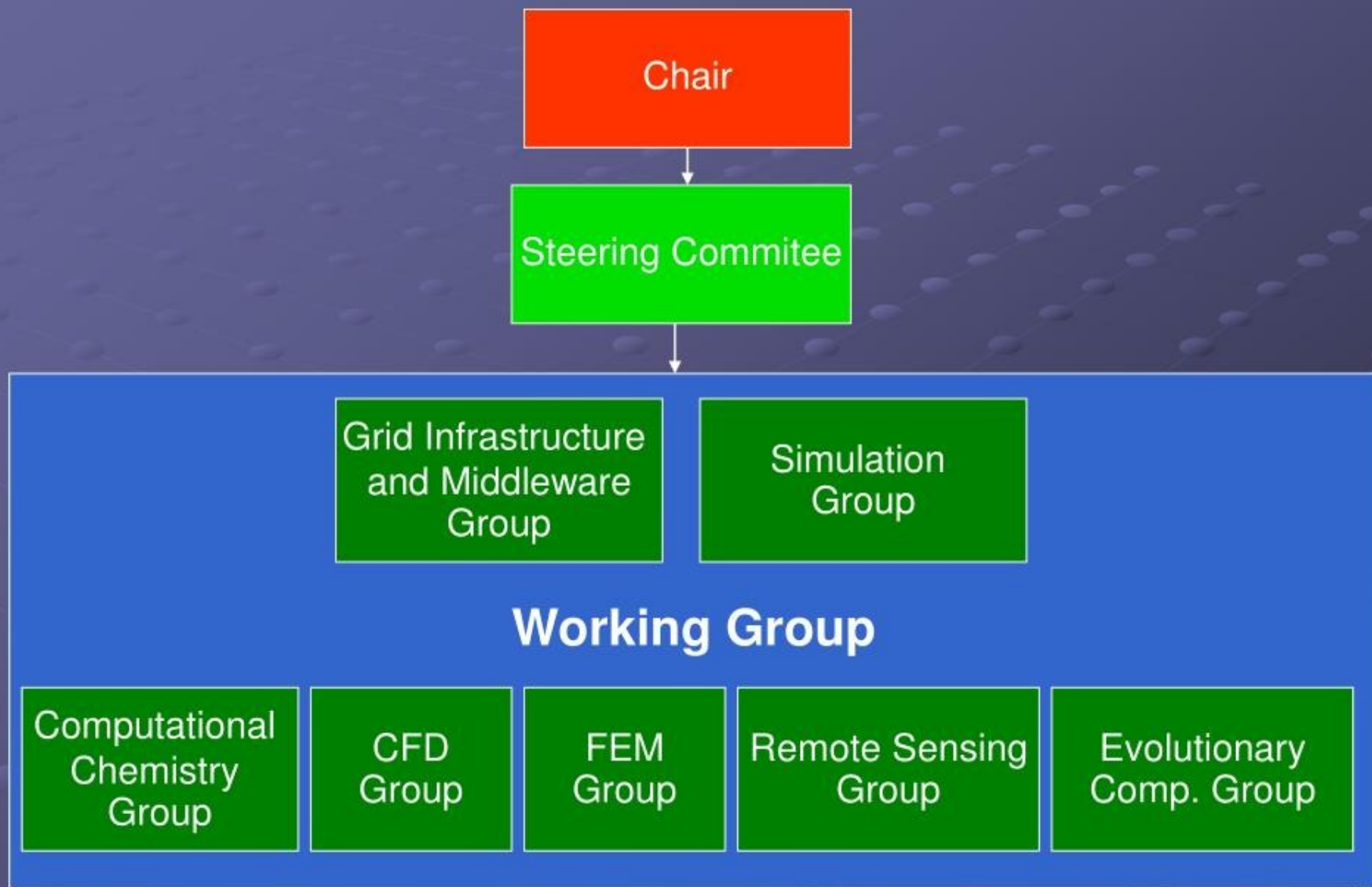
- National Electronics and Computing Technology



Goal

- Create a grid computing infrastructure for Thai researchers
- Stimulate the deployment of Grid Computing Technology
- Build a collaborative Research Network among Thai researchers
- Act as a focal point for international grid collaboration

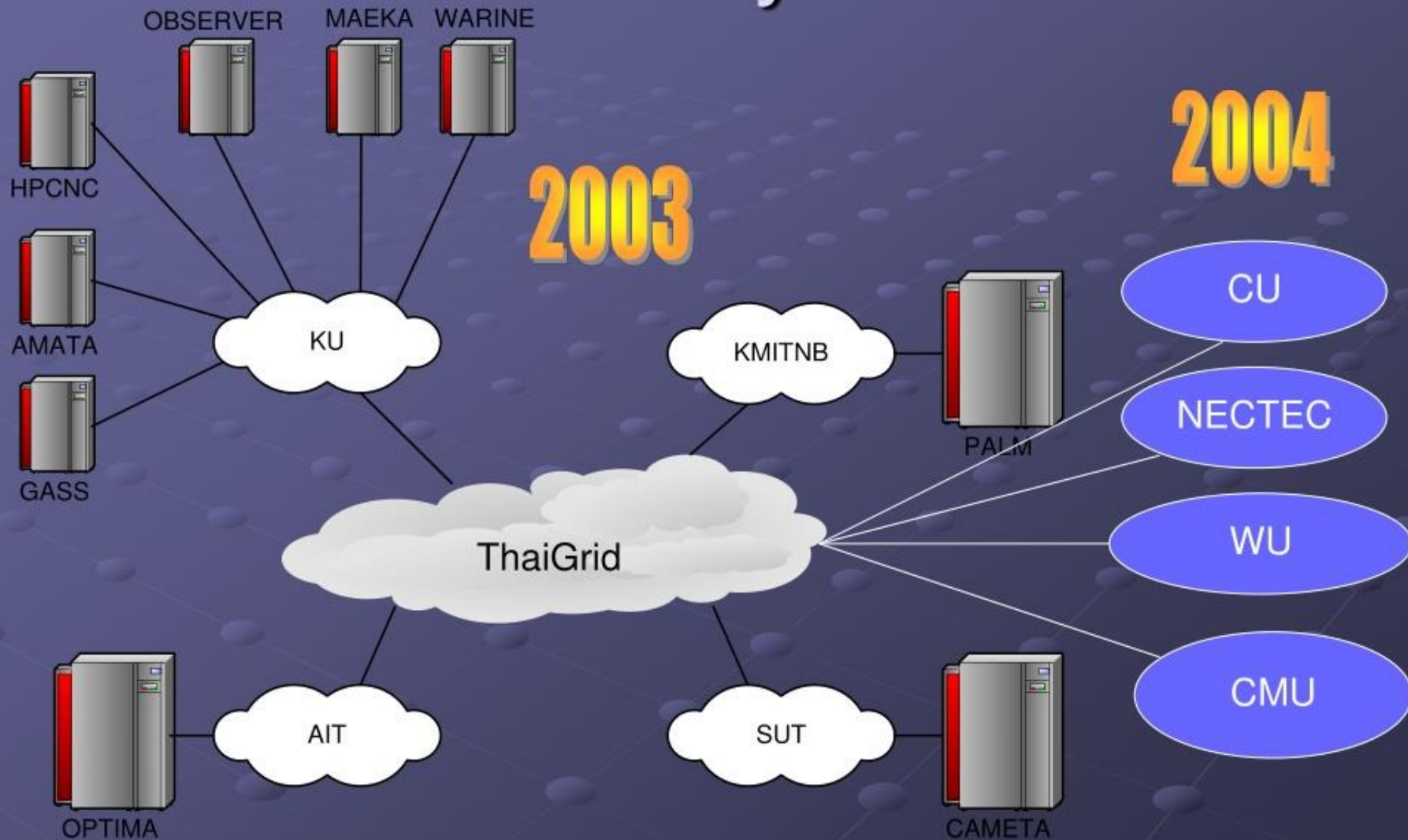
Organization



Activities

- Building ThaiGrid Testbed
- Research
 - Tools
 - Applications
- International Collaboration
 - ApGrid
 - PRAGMA
 - APAN

ThaiGrid System

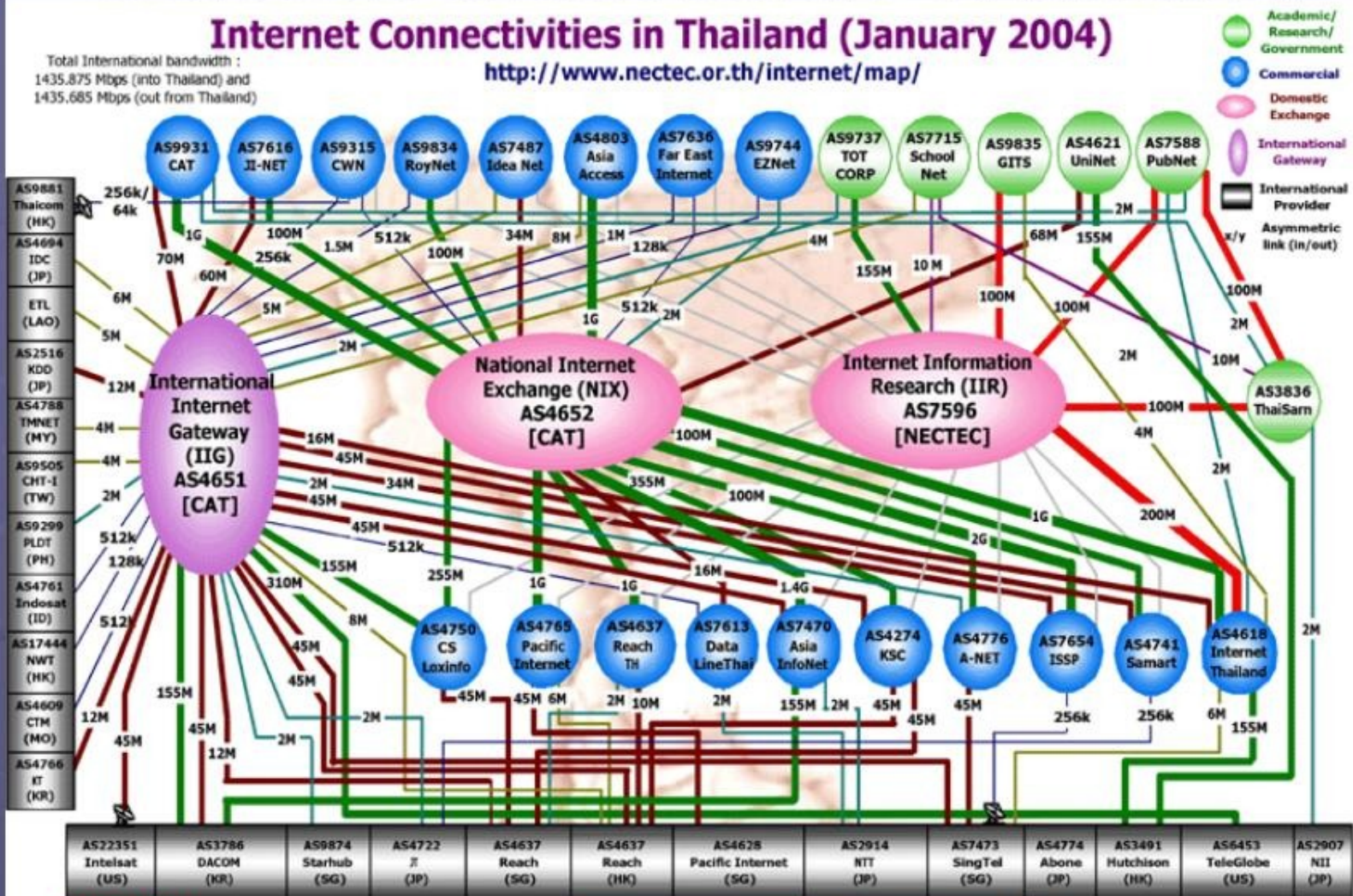


Current Internet in Thailand

Internet Connectivities in Thailand (January 2004)

Total International bandwidth :
1435.875 Mbps (into Thailand) and
1435.685 Mbps (out from Thailand)

<http://www.nectec.or.th/internet/map/>



DISCLAIMER

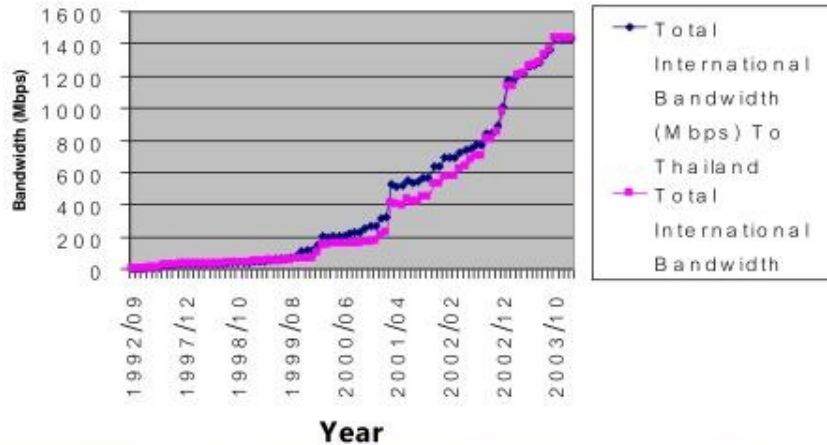
Chart Date: 2004-01-01

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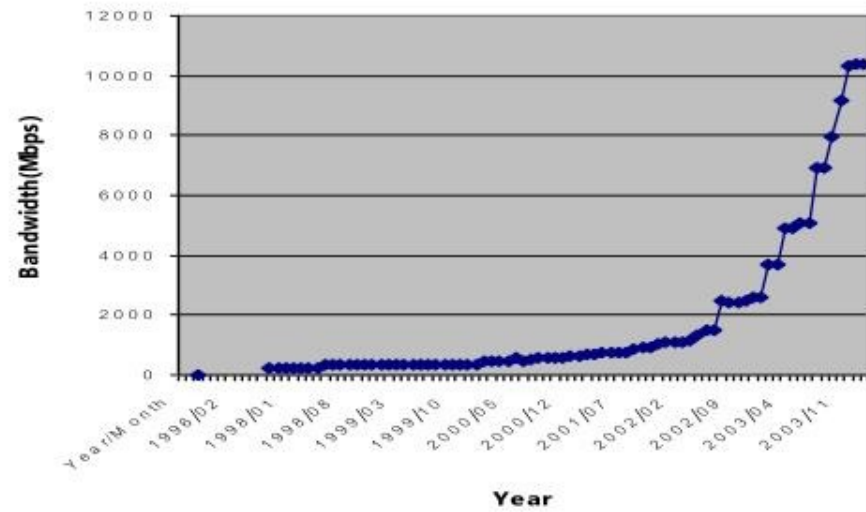


Bandwidth

Thailand's International Bandwidth



Thailand's Internal Bandwidth



ThaiGrid core Network

● THAISARN

- 2 Mbps link supported by NECTEC
 - NECTEC- KMITNB
 - NECTEC- SUT
- NECTEC-KU ATM 155Mbps

● UNINET

- KU-UNINET 155 Mbps
- AIT-UNINET 155Mbps

● UNINET-Internet 2 45Mbps

Resources (2003- First Part of 2004)

● KU

- MAEKA
 - 32 nodes dual processors AMD Opteron 1.4Ghz, 3GB Mem 80Gb HDD, Gigabit Ethernet
- GASS
 - 6 nodes DUAL AMD Athlon MP1800+, 1GB RAM 80 GB HDD
 - Gigabit Ethernet
- WARINE
 - 16 nodes Celeron 2Ghz , 512 Mb RAM 80GB HDD, Fast Ethernet
- AMATA
 - 14 nodes AMD 1GHZ 512 MB 40GB Fast Ethernet Myrinet (6 nodes)
- HPCNC
 - 1 nodes ATHLON 1800+ , 512 MB RAM, 80GB HDD
- OBSERVER
 - 1 nodes Athlon 1800+ 512 MB RAM, 80 GB HDD

● KMITNB

- PALM
 - 16 nodes Pentium 4 2GHz 512 MB RAM, Fast Ethernet
- Enqueue
 - 9 nodes Dual AMD 2.2GHz, 1GB RAM 32GFLOPs, 2G Myrinet

● AIT

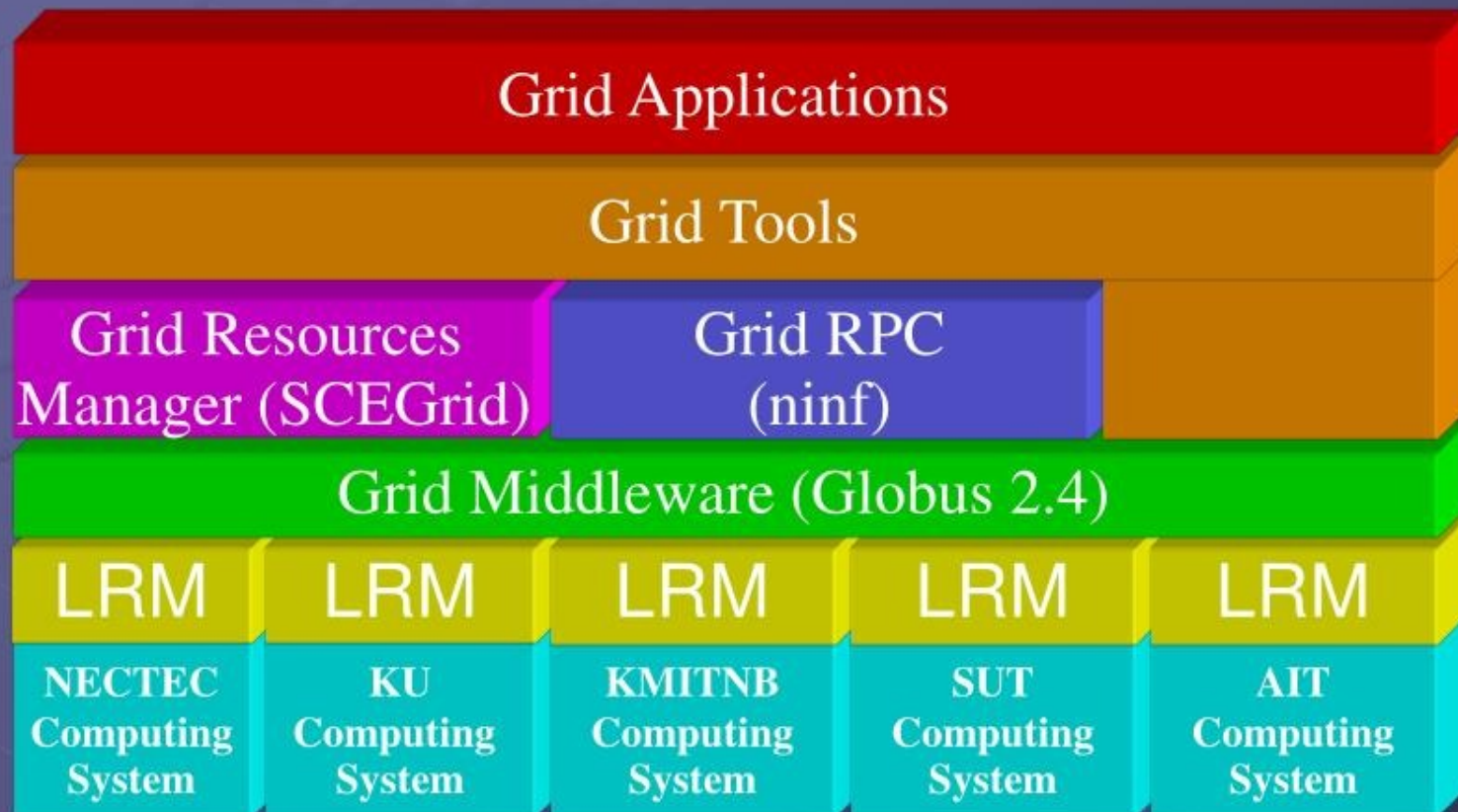
- OPTIMA
 - 8 nodes Athlon XP1800+
 - Fast Ethernet

● SUT

- CAMETA
 - 16 nodes Athlon XP1800+
 - Fast Ethernet

Total of 148 processors on ThaiGrid

ThaiGrid Software Architecture



LRM=Local Resources Manager

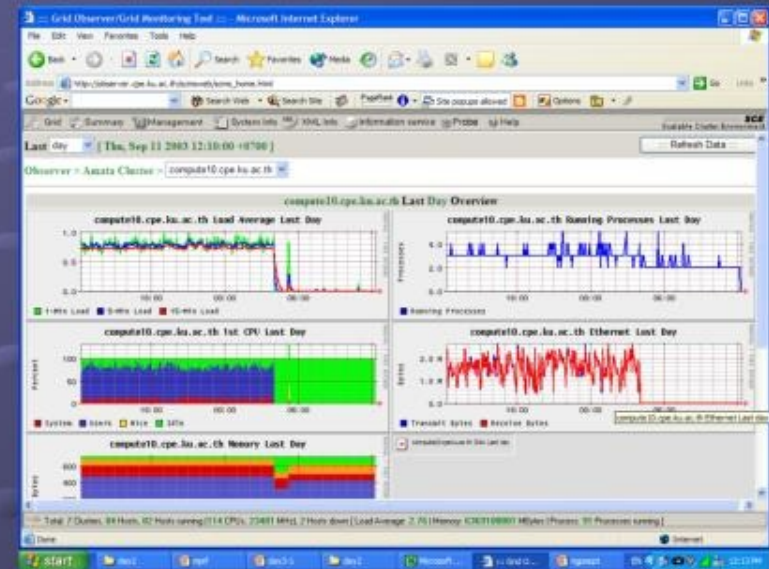
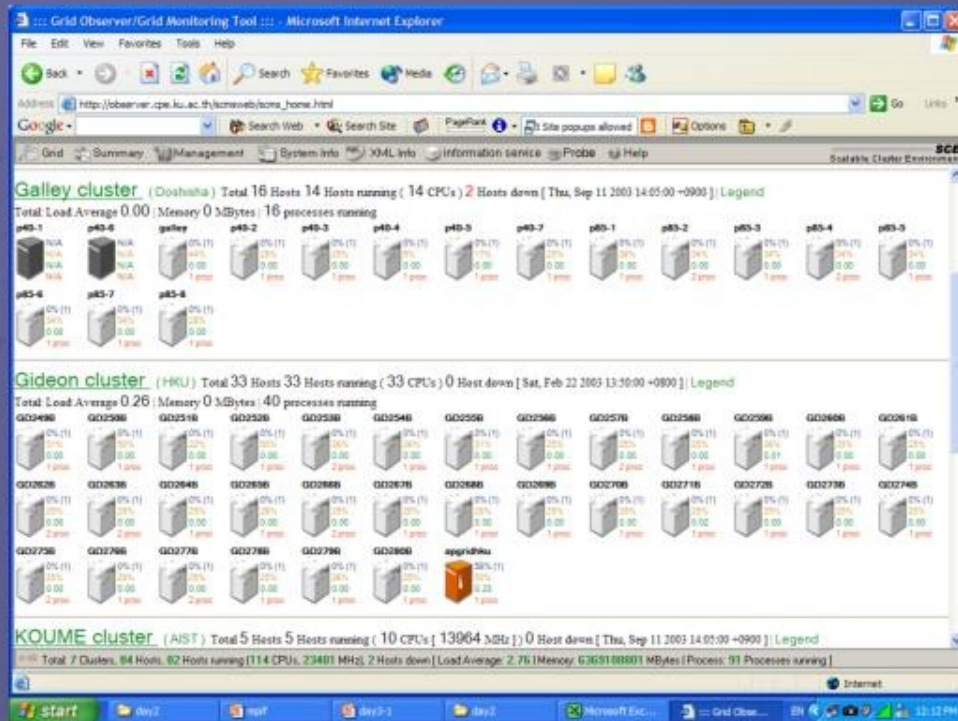
Software

- Local Resources Management
 - Condor
 - SQMS (KU)
 - SGE (Planned)
- Middleware
 - Globus 2.4
- Grid Level Resource Management
 - SCEGrid Scheduler (KU)
- Data Grid
 - Gfarm Data Grid (AIST)
- Grid Programming Environment
 - Ninf GridRPC (AIST)
 - MPICH-G2
- Tools
 - SCMSweb Monitoring (KU)

Tools Development

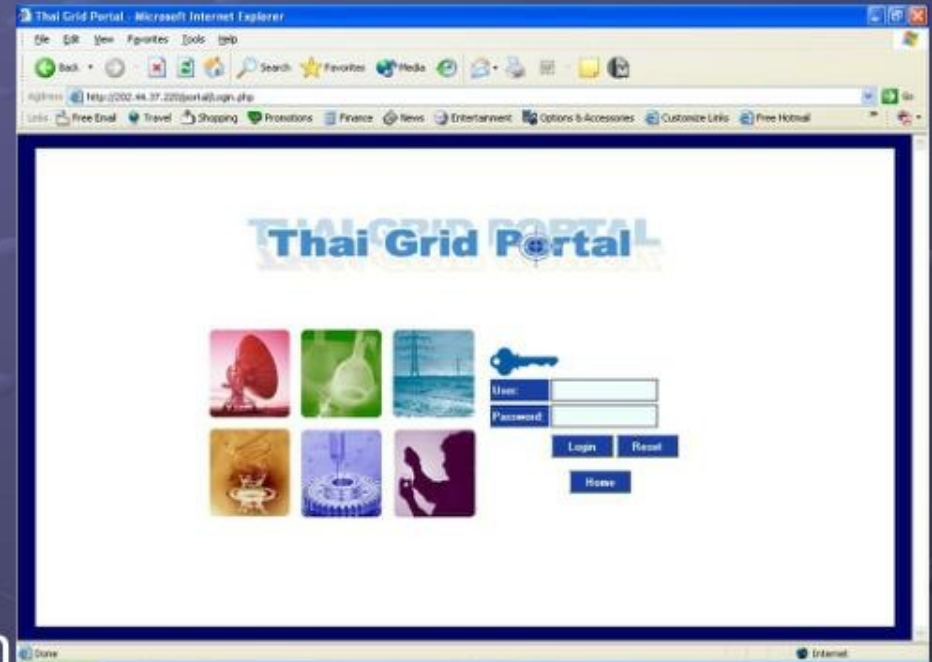
- OpenSCE: Cluster software Tools and Middleware (KU)
 - MPview – MPI program visualization
 - MPITH – Quick and simple MPI runtime for cluster and grid
 - SQMS – Batch scheduler for cluster
 - SCMS/ SCMSWEB cluster management tool
- ThaiGrid Portal (KMITNB)
- HypersimGrid Simulator for Grid design (KU)

SCMS Web Monitoring



ThaiGrid Portal

- Data Manage.
- Web-base Compilers.
- Jobs Submission.
- Jobs Manage.
- Resources Monitoring.
- Automatic and Manual generate RSL.
- User Management.
- Portal systems configuration.



ThaiGrid Portal

- Portal are centralize of grid computing.
- Middle tier between grid servers and grid users.
- Developed on Web technology.
- Allocate the appropriate resources.
- Use XML for standard document.
- Use web account only, Portal CA.

ThaiGrid Portal Jobs function

- Scheduler Job.
- Generate RSL files.
- Supports serial and parallel jobs.

Thai Grid Portal - Microsoft Internet Explorer

Address: http://202.44.37.220/~pao/ThaiGrid/Job_RSL/submit_selected_data.php

Thai Grid Portal

Selected Options

Username: pao

Number of Processes: 1

Job Type: Parallel Job

File Name: Hello.c.out

RSL Generate: Automatic

Running On: 0 : 00

Please choose an email, if you want to receive output file by email.

Email Address:

Portal - Microsoft Internet Explorer

Address: http://202.44.37.220/portal/admin_data/job_manage.php

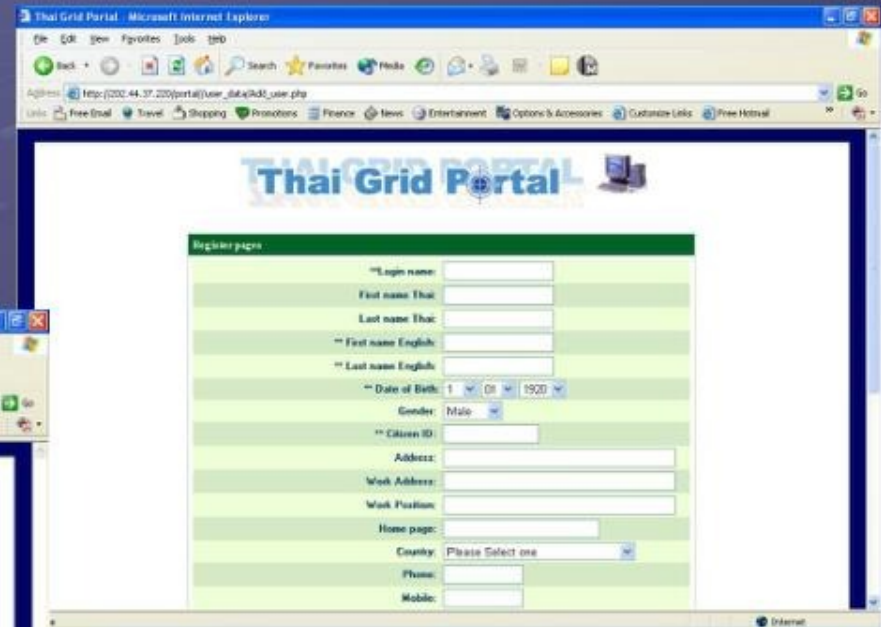
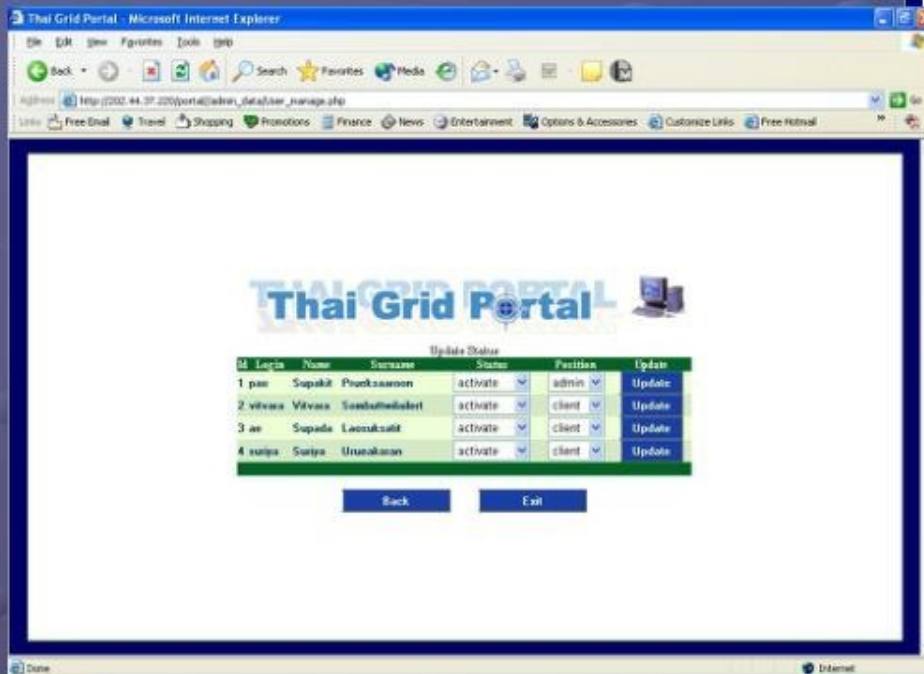
Thai Grid Portal

Old Status

Server	Job ID	Job Status	Type	Machine	Manage
202.44.37.220	job25049	queued	Null	Null	Manage
202.44.37.220	job6739	queued	Null	Null	Manage
202.44.37.220	job25722	queued	Null	Null	Manage
202.44.37.220	job6364	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job6364	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job1563	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job1563	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job18372	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job18372	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job5342	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job5342	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job12177	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job12177	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job12177	running	machine	petonia.ee.ku.ac.th	Manage
202.44.37.220	job12177	running	machine	duhla.ee.ku.ac.th	Manage
202.44.37.220	job6605	queued	Null	Null	Manage
202.44.37.220	job6602	queued	Null	Null	Manage

Portal User function

- Registration.
- Activate/deactivate account.
- Edits user information.



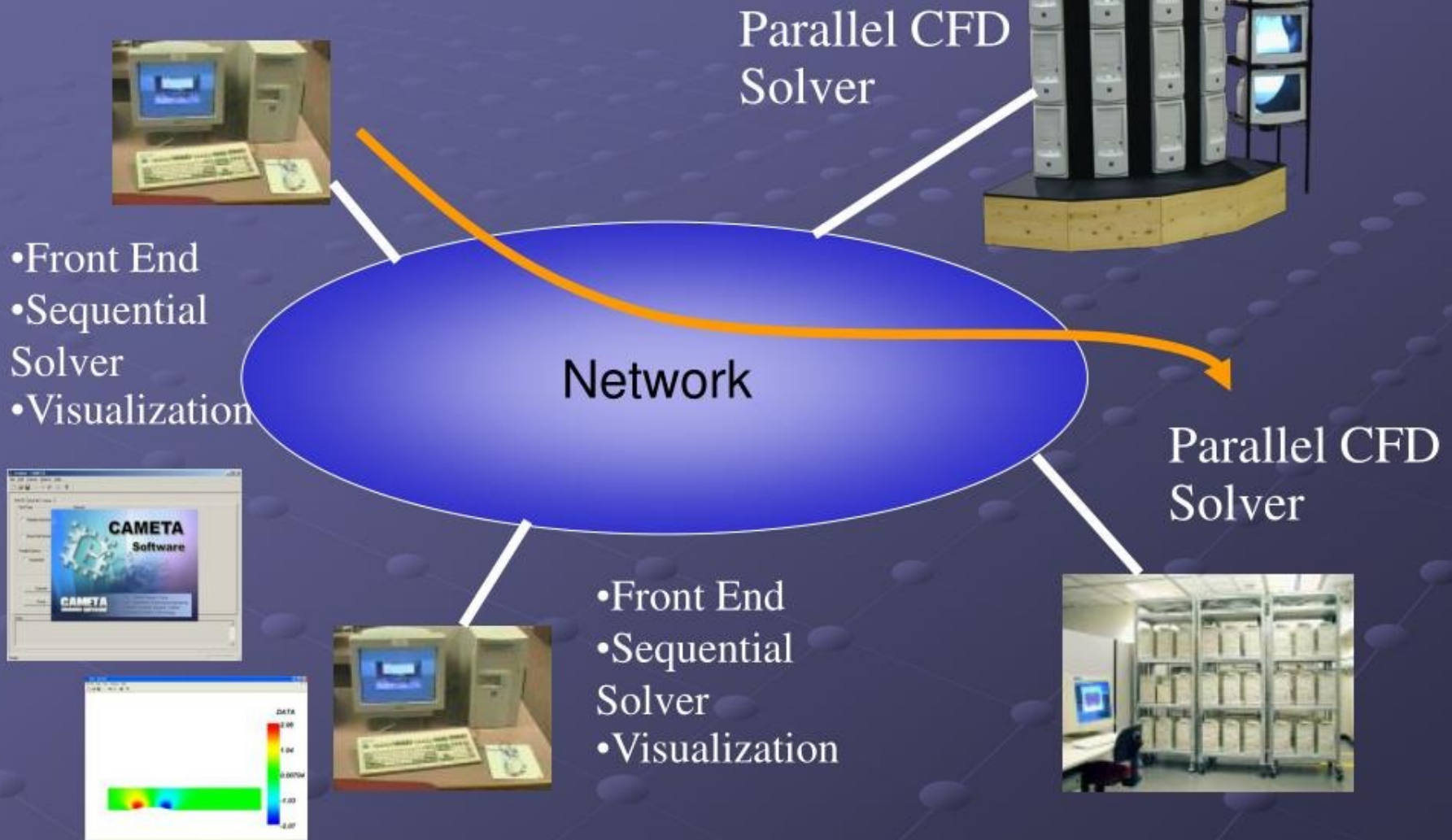
Application

- Computational Fluid Dynamics
- Simulation
 - Scheduling
 - PGA Pack
- Computational Chemistry
 - GAMESS (General Atomic and Molecular Electronic Structure System)
- FEM in High Voltage Insulator
- Evolutionary Computing

Clean Room Project

- Member: KU, SUT
- Goal: study clean room using CFD
 - Three-Dimensional Turbulence Problem
 - Heat & Mass Transfer
- Using:
 - Finite volume, Multigrid, Parallel computing
- Solution: Grid is used to
 - Provided uniform security mechanism across the cluster computing environment
 - Provide mechanism for large scale data access
- Tools
 - Globus , MPICH
 - Grid RPC (ninf, netsolve)
 - Gfarm data grid

Software Structure



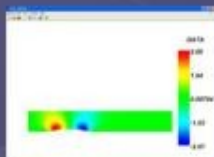
Operation

Windows

User Input
Problem

gridview

Scview



Grid

ACI

Linux Cluster

SQMS

Parallel Solver

Simulation

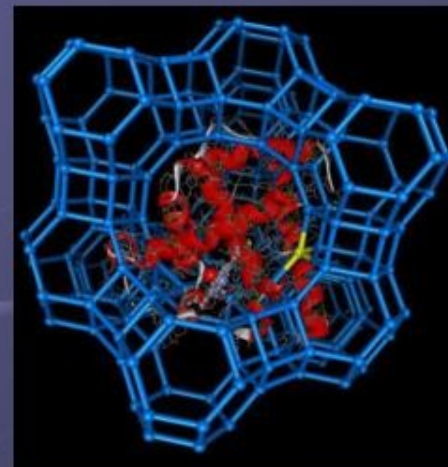
- Many simulation and optimization problem can utilized grid and cluster well
- Parametric applications is perfect for grid
- Simulation job on the ThaiGrid
 - Genetic algorithm for optimization problem using PGAPack
 - Grid simulation (HyperGridSim)

Solution

- Running on cluster using batch scheduler
- Deploy over Grid
 - Using SCE/Grid scheduler
- Tools
 - Globus
 - SCE/Grid
 - SQMS, SGE, OPENPBS

Computational Chemistry

- Laboratory for Computational and Applied Chemistry (LCAC), KU.
- Research
 - Zeolite Chemistry & Catalysis
 - Surface Structure & Reactivity of Advanced Materials
 - calculate molecular structures and properties of HIV-1 inhibitors in the class of non-nucleoside derivatives and to create quantitative structure-activity relationships (QSAR) model, based on both classical and 3-Dimensional QSAR.



Solution

- Running GAMESS on cluster (currently)
- Deploy GAMESS over Grid
 - Using SCE/Grid scheduler
- Tools
 - Globus
 - SQMS/Grid
 - SQMS, SGE, OPENPBS

Remote Sensing

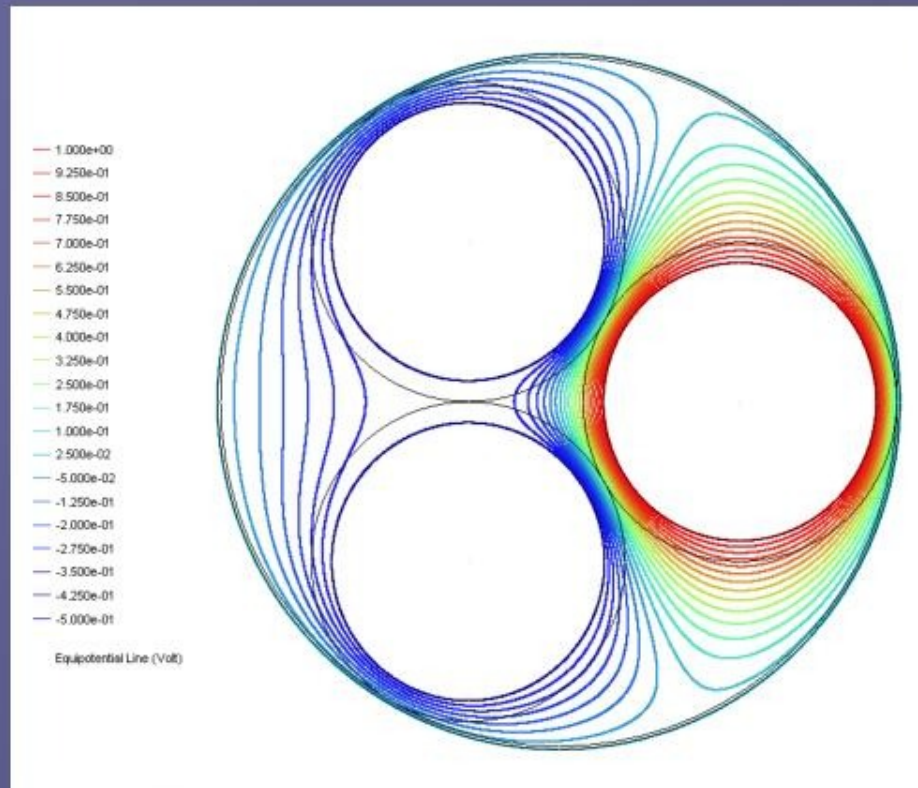
Star Project (KU/AIT)

- Deploy cluster and grid for remote sensing application
- Analysis of the impact of irrigation system using image processing and genetics algorithm

● Approach

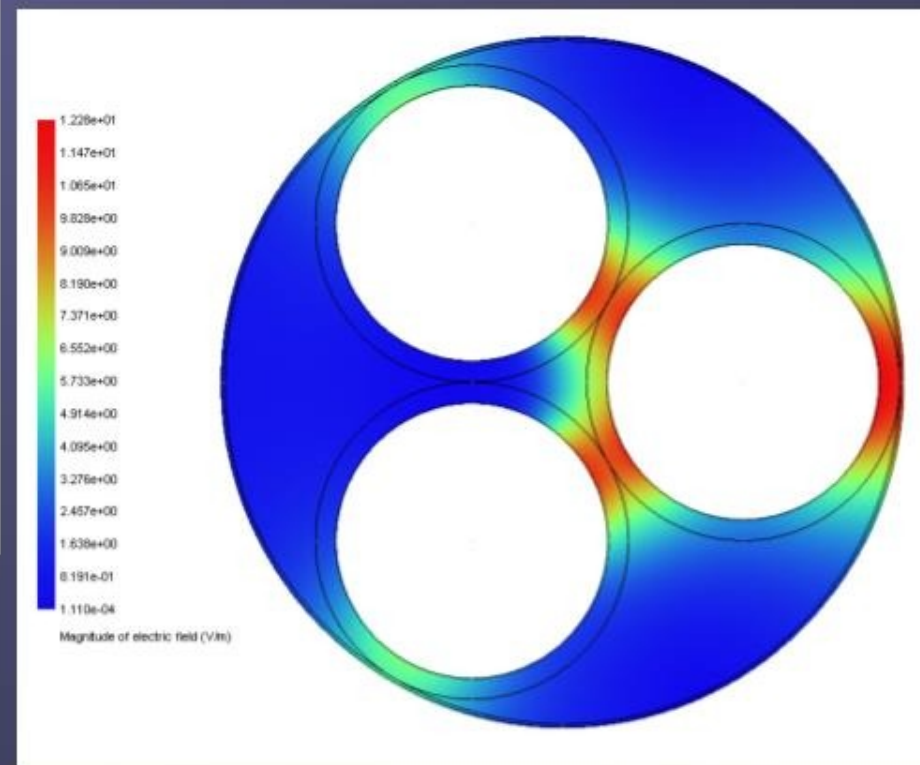
- Using gridrpc for parallelization
- Using batch scheduler for GA simulation

Parallel Electric field Calculation : High Performance Library Integrated Approach

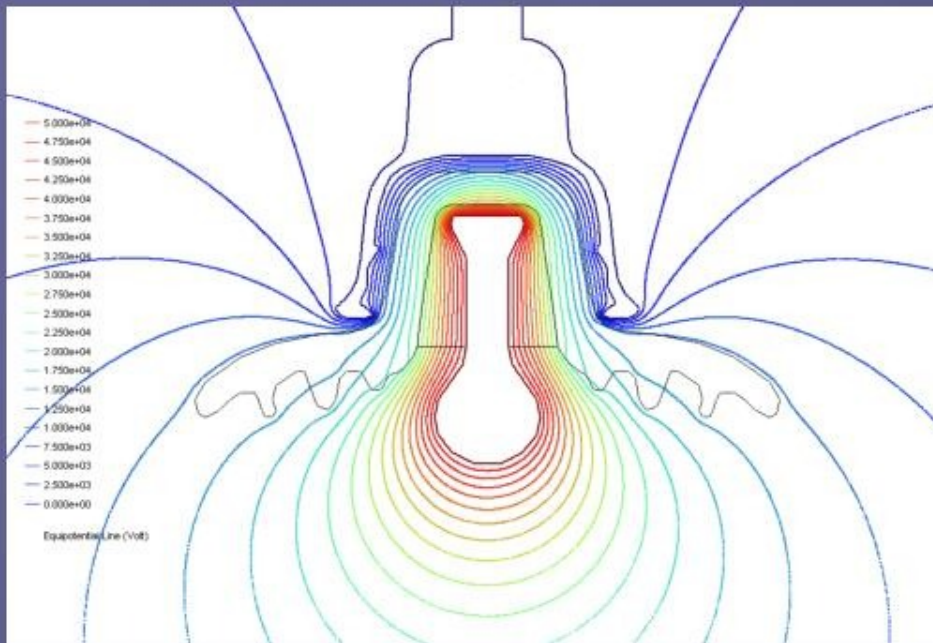


948,018 nodes and
1,887,408 elements

Analyze electrical stress on Three Phases Power Cable

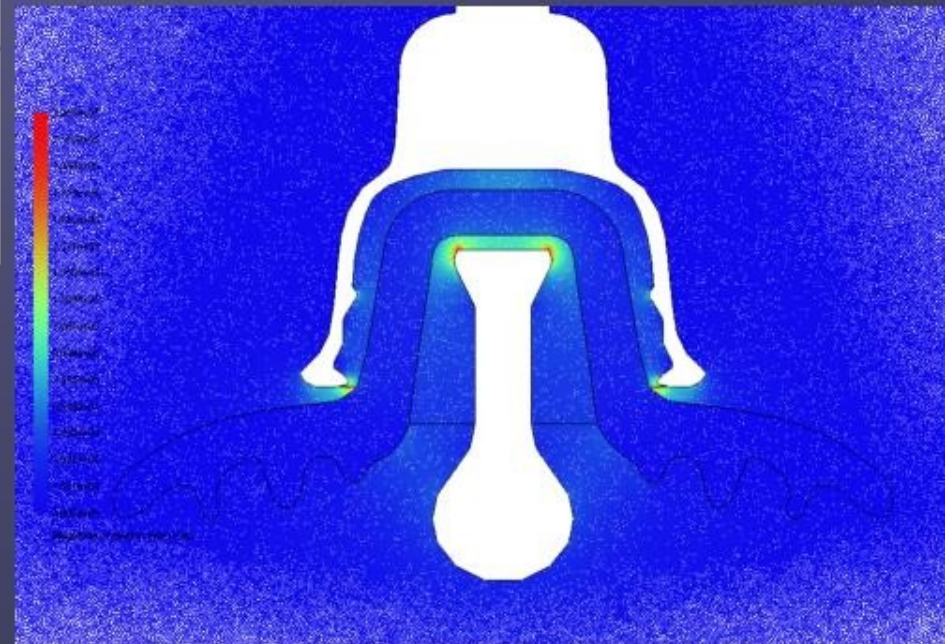


Parallel Electric field Calculation : High Performance Library Integrated Approach



680,583 nodes and
1,357,963 elements

Analyze electrical stress on
High Voltage Insulator



Evolutionary Computation: Theories and Applications in Engineering, Biology, and Medicine

- **Investigators:** Nachol Chaiyaratana and Vara Varavithya



Evolutionary computation concerns theories and applications of biologically inspired algorithms. Similar to biological systems, the solutions generated by these algorithms are allowed to emerge or change through the processes of evolution or adaptation as guided by external stimuli. Our research interests cover both theories and applications of various techniques including genetic algorithms, genetic programming and ant colony system algorithms.

Theory

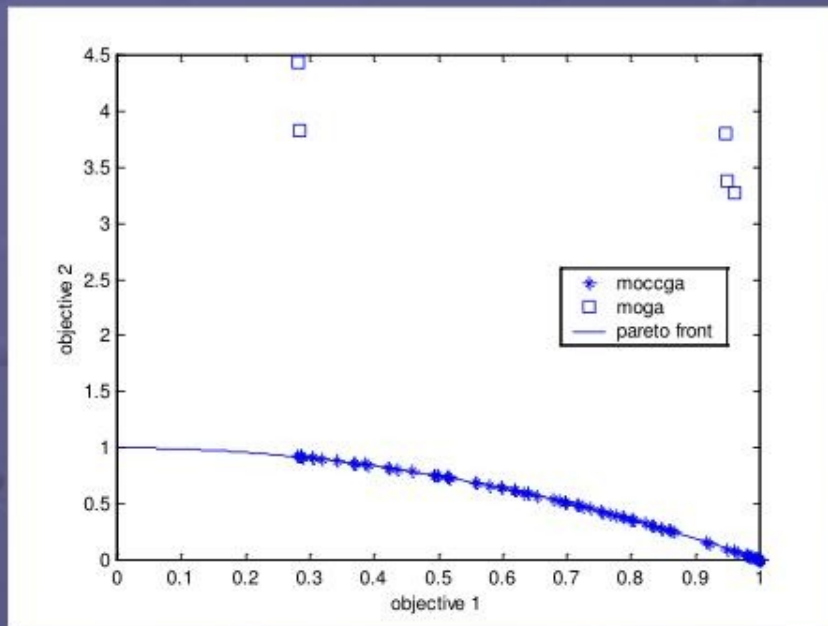
1. Multi-Objective Co-Operative Co-Evolutionary Genetic Algorithm
2. Diversity Control in a Multi-Objective Genetic Algorithm

Application

1. Wireless LAN Access Point Placement using a Multi-Objective Genetic Algorithm
2. DNA Fragment Assembly using an Ant Colony System Algorithm
3. Thalassaemic Patient Classification using a Neural Network and Genetic Programming

Multi-Objective Co-Operative Co-Evolutionary Genetic Algorithm

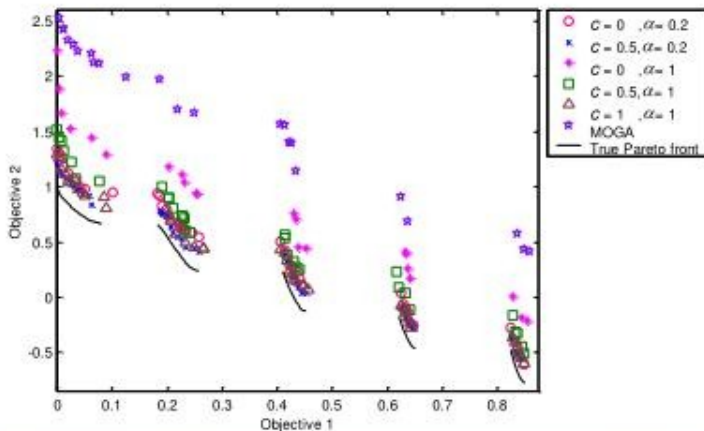
- **Investigators:** Nuttavut Keerativuttitumrong, Nachol Chaiyaratana and Vara Varavithya



- Integration between two types of genetic algorithm: a multi-objective genetic algorithm (MOGA) and a co-operative co-evolutionary genetic algorithm (CCGA)
- Improve the performance of the MOGA by adding the co-operative co-evolutionary effect to the search mechanisms employed by the MOGA
- In overall the MOCCGA is superior to the MOGA in terms of the variety in solutions generated and the closeness of solutions to the true Pareto-optimal solutions
- With the use of an 8-node cluster, the speed-up of 2.64 to 4.8 can be achieved for the test problems

Diversity Control in a Multi-Objective Genetic Algorithm

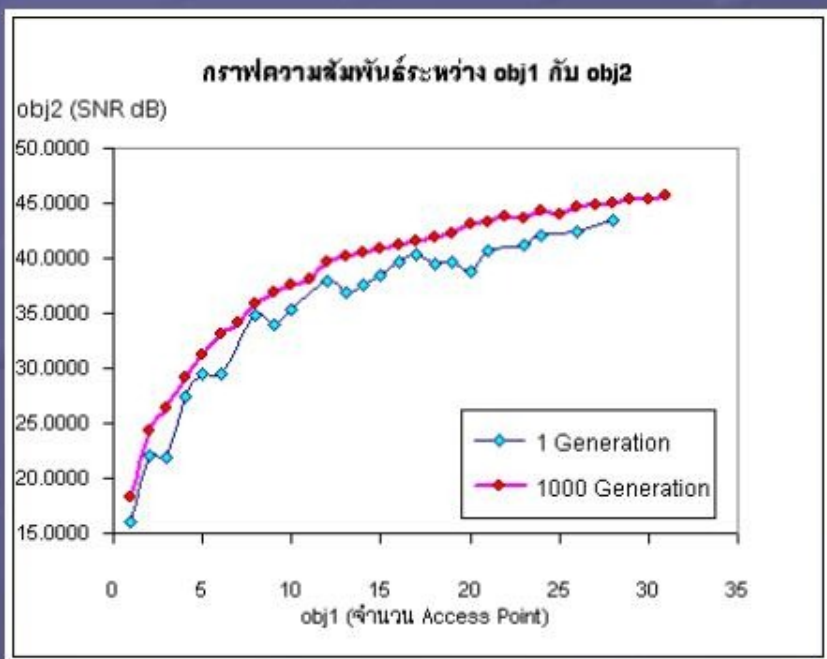
- **Investigators:** Nuntapon Sangkawelert and Nachol Chaiyaratana



- The diversity control operator used is based on the one developed for a diversity control oriented genetic algorithm (DCGA).
- The performance comparison between multi-objective genetic algorithms with and without diversity control is explored where different benchmark problems with specific multi-objective characteristics are utilised.
- The results indicate that the use of diversity control with specific parameter settings promotes the emergence of multi-objective solutions that are close to the true Pareto optimal solutions while maintaining a uniform distribution of the solutions along the Pareto front.

Wireless LAN Access Point Placement using a Multi-Objective Genetic Algorithm

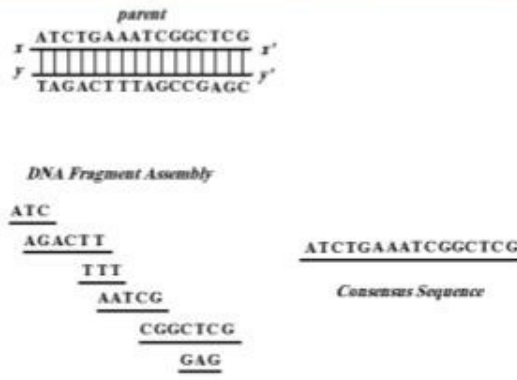
- **Investigators:** Kotchakorn Maksuriwong, Vara Varavithya and Nachol Chaiyaratana



- The aim is to maximise signal coverage over an interested area.
- The decision variables are derived from the locations of the access points.
- The objectives consist of the number of access points and the average SNR over the whole area.
- The MOGA is capable of generating a placement result which is superior to that produced using standard placement techniques.
- Multiple optimal placement configurations for different numbers of access points can be obtained from a single run of the MOGA.

DNA Fragment Assembly using an Ant Colony System Algorithm

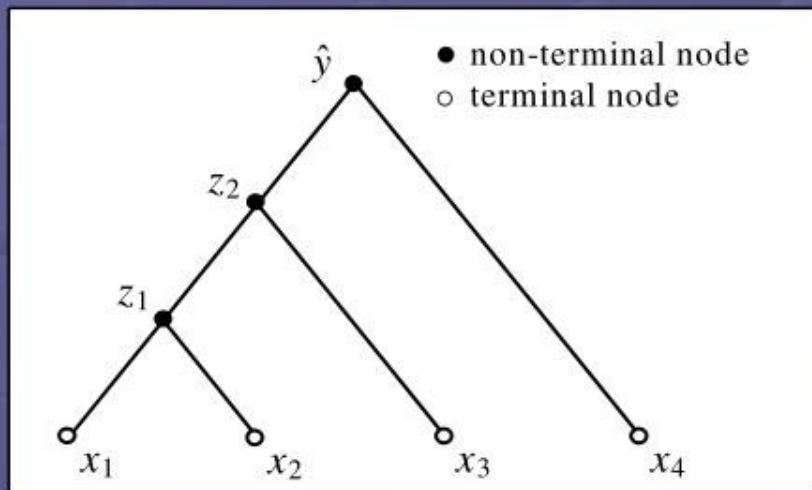
- **Investigators:** Prakrit Meksangsouy and Nachol Chaiyaratana



- The aim is to find the right order and orientation of each fragment in the fragment ordering sequence that leads to the formation of a consensus sequence.
- An asymmetric ordering representation is proposed where a path co-operatively generated by all ants in the colony represents the search solution.
- The optimality of the fragment layout is obtained from the sum of overlap scores calculated for each pair of consecutive fragments.
- The ant colony system algorithm outperforms the nearest neighbour heuristic algorithm when multiple-contig problems are considered.

Thalassemic Patient Classification using a Neural Network and Genetic Programming

- **Investigators:** Waranyu Wongseree and Nachol Chaiyaratana



- Using a genetic programming (GP) system called STROGANOFF and a multilayer perceptron in thalassemic patient classification
- The problem covers the test samples from normal subjects and that from different types of thalassemic patient and thalassemic trait.
- The characteristics of red blood cell, reticulocyte and blood platelet are used as input.
- The performance of the GP-generated classification trees is approximately equal to that of the multilayer perceptrons.
- The structure of the classification trees reveals that the characteristics of blood platelet have no effects on the classification performance.

Related Project

- Thai e-science project

- New project funded in 2003 (3 Million)
- Application oriented project

- Current members

- Computational Chemistry Unit Cell, Department of Chemistry, Chulalongkorn University
- Department of Computer Engineering, Chulalongkorn University
- HPCNC, Kasetsart University

- Contact: <http://www.thai-escience.net/>

- Dr. Prabhas Chongstitvatana (Associate Professor, Intelligent System Lab, Department of Computer Engineering, Chulalongkorn University) prabhas.c@chula.ac.th

Conclusion

- Grid is a promising technology but
 - Lack manpower and expertise
 - Difficult to setup , steep learning curve
- The awareness of Grid and E-science in Thailand is still at the very beginning
- There is a need to
 - Build larger community, focus more on education and out-reach program
 - Build strong testbed first
 - Find killer applications

Future Plan

- Building easy to use and stable environment
- Attract more user and more applications
 - Bioinformatics
 - Nanotechnology
- Find new area to deploy grid technology
- Education, technology transfer



The end